

Solution to Homework 4

Total 30 points

Problem 1. (8 points) Suppose f is a convex and differentiable function, $\mathcal{C}$ is a closed convex set. Show that

$$\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x}) \iff \langle -\nabla f(\mathbf{x}^*), \mathbf{z} - \mathbf{x}^* \rangle \leq 0, \forall \mathbf{z} \in \mathcal{C}.$$

Solution.

$\Rightarrow$: We prove by contradiction. Suppose there exists $\mathbf{y} \in \mathcal{C}$ such that $\langle \nabla f(\mathbf{x}^*), \mathbf{y} - \mathbf{x}^* \rangle < 0$.

Consider $\mathbf{z}(t) = t\mathbf{y} + (1-t)\mathbf{x}^*$ where $t \in [0, 1]$ is a parameter. Since $\mathbf{x}^*, \mathbf{y} \in \mathcal{C}$, we know $\mathbf{z}(t) \in \mathcal{C}$. Since

$$\left. \frac{d}{dt} f(\mathbf{z}(t)) \right|_{t=0} = \langle \nabla f(\mathbf{x}^* + 0 \cdot (\mathbf{y} - \mathbf{x}^*)), \mathbf{y} - \mathbf{x}^* \rangle = \langle \nabla f(\mathbf{x}^*), \mathbf{y} - \mathbf{x}^* \rangle < 0,$$

we can get for small positive t , $f(\mathbf{z}(t)) < f(\mathbf{x}^*)$.

$\Leftarrow$: Since $f(\mathbf{x})$ is convex, we have

$$f(\mathbf{x}^*) \leq f(\mathbf{z}) + \langle -\nabla f(\mathbf{x}^*), \mathbf{z} - \mathbf{x}^* \rangle \leq f(\mathbf{z}), \forall \mathbf{z} \in \mathcal{C}.$$

Problem 2. (5 points) Let $\mathcal{C} \in \mathbb{R}^d$ be a nonempty closed and convex set, and let f be a strongly convex function over $\mathcal{C}$. Prove that f has a unique minimizer $\mathbf{x}^*$ over $\mathcal{C}$.

Solution. Suppose there are two minimizer $\mathbf{x}^*$ and $\mathbf{y}^*$. Since f is strongly convex, we have

$$f(\mathbf{y}^*) \geq f(\mathbf{x}^*) + \langle \nabla f(\mathbf{x}^*), \mathbf{y}^* - \mathbf{x}^* \rangle + \frac{\mu}{2} \|\mathbf{x}^* - \mathbf{y}^*\|_2^2.$$

Since $\mathbf{x}^*$ is a minimizer, according to the problem 1, we know $\langle -\nabla f(\mathbf{x}^*), \mathbf{y}^* - \mathbf{x}^* \rangle \leq 0$. Thus

$$\frac{\mu}{2} \|\mathbf{x}^* - \mathbf{y}^*\|_2^2 \leq \langle -\nabla f(\mathbf{x}^*), \mathbf{y}^* - \mathbf{x}^* \rangle + f(\mathbf{y}^*) - f(\mathbf{x}^*) \leq 0.$$

which means $\mathbf{x}^* = \mathbf{y}^*$.

Problem 3. (5 points) Consider the projected gradient descent algorithm introduced in the class. Suppose that for some iteration t , $\mathbf{x}_{t+1} = \mathbf{x}_t$. Prove that in this case, $\mathbf{x}_t$ is a minimizer of the convex objective function f over the closed and convex set $\mathcal{C}$.

Solution. Let $\mathbf{y}_{t+1} = \mathbf{x}_t - \eta_t \nabla f(\mathbf{x}_t)$, then we know $\mathbf{x}_{t+1} = \mathcal{P}_{\mathcal{C}}(\mathbf{y}_{t+1})$. According to property 1 in page 8 of the slides, we have

$$\langle \mathbf{x} - \mathbf{x}_{t+1}, \mathbf{y}_{t+1} - \mathbf{x}_{t+1} \rangle \leq 0, \forall \mathbf{x} \in \mathcal{C}.$$

Since $\mathbf{x}_{t+1} = \mathbf{x}_t$ and $\mathbf{y}_{t+1} - \mathbf{x}_t = -\eta_t \nabla f(\mathbf{x}_t)$, we have

$$\langle \mathbf{x} - \mathbf{x}_t, -\eta_t \nabla f(\mathbf{x}_t) \rangle \leq 0, \forall \mathbf{x} \in \mathcal{C}.$$

According to the problem 1, we know $\mathbf{x}_t \in \arg \min_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x})$.

Problem 4. (12 points) Suppose f is a L -smooth and μ -strongly convex function, $\mathcal{C}$ is a closed and convex set.

- Let $T(\mathbf{x}) = \mathcal{P}_{\mathcal{C}}(\mathbf{x} - \frac{1}{L} \nabla f(\mathbf{x}))$ and $g_{\mathcal{C}}(\mathbf{x}) = L(\mathbf{x} - T(\mathbf{x}))$, show that

$$\langle g_{\mathcal{C}}(\mathbf{x}), \mathbf{x} - \mathbf{x}^* \rangle \geq \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|_2^2 + \frac{1}{2L} \|g_{\mathcal{C}}(\mathbf{x})\|_2^2.$$

- Show the projected gradient descent with $\eta = 1/L$ has the following convergence rate:

$$\|\mathbf{x}_t - \mathbf{x}^*\|_2^2 \leq (1 - \frac{\mu}{L})^t \|\mathbf{x}_0 - \mathbf{x}^*\|_2^2.$$

Solution.

- By smoothness and strong convexity, we have

$$\begin{aligned} 0 &\leq f(T(\mathbf{x})) - f(\mathbf{x}^*) = f(T(\mathbf{x})) - f(\mathbf{x}) + f(\mathbf{x}) - f(\mathbf{x}^*) \\ &\leq \langle \nabla f(\mathbf{x}), T(\mathbf{x}) - \mathbf{x} \rangle + \frac{L}{2} \|T(\mathbf{x}) - \mathbf{x}\|_2^2 + \langle \nabla f(\mathbf{x}), \mathbf{x} - \mathbf{x}^* \rangle - \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|_2^2 \\ &= \langle \nabla f(\mathbf{x}), T(\mathbf{x}) - \mathbf{x}^* \rangle + \frac{1}{2L} \|g_{\mathcal{C}}(\mathbf{x})\|_2^2 - \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|_2^2 \end{aligned}$$

In the class, we have shown that $\langle \nabla f(\mathbf{x}), T(\mathbf{x}) - \mathbf{x}^* \rangle \leq \langle g_{\mathcal{C}}(\mathbf{x}), T(\mathbf{x}) - \mathbf{x}^* \rangle$. Thus we have

$$\langle g_{\mathcal{C}}(\mathbf{x}), \mathbf{x} - \mathbf{x}^* \rangle = \langle g_{\mathcal{C}}(\mathbf{x}), T(\mathbf{x}) - \mathbf{x}^* \rangle + \frac{1}{L} \|g_{\mathcal{C}}(\mathbf{x})\|_2^2 \geq \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|_2^2 + \frac{1}{2L} \|g_{\mathcal{C}}(\mathbf{x})\|_2^2.$$

- We can get

$$\begin{aligned} \|\mathbf{x}_{t+1} - \mathbf{x}^*\|_2^2 &= \|\mathbf{x}_{t+1} - \mathbf{x}_t + \mathbf{x}_t - \mathbf{x}^*\|_2^2 \\ &= \|T(\mathbf{x}_t) - \mathbf{x}_t\|_2^2 + \|\mathbf{x}_t - \mathbf{x}^*\|_2^2 + 2\langle T(\mathbf{x}_t) - \mathbf{x}_t, \mathbf{x}_t - \mathbf{x}^* \rangle \\ &= \frac{1}{L^2} \|g_{\mathcal{C}}(\mathbf{x}_t)\|_2^2 + \|\mathbf{x}_t - \mathbf{x}^*\|_2^2 - \frac{2}{L} \langle g_{\mathcal{C}}(\mathbf{x}), \mathbf{x} - \mathbf{x}^* \rangle \\ &\leq (1 - \frac{\mu}{L}) \|\mathbf{x}_t - \mathbf{x}^*\|_2^2. \end{aligned}$$